

Matter, Antimatter and Dark Matter

A Unified Perspective from Relational Geometry

Draft v1

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Manuscript status — Draft v1. This note presents the definitive synthesis of the relational geometry framework applied to matter–antimatter asymmetry and dark matter. All claims are labelled according to the taxonomy of Table 1.

Key results:

- The propagation invariant $R_V(n)\omega_V(n) = c$ is derived as a necessary algebraic consistency condition, not merely a numerical coincidence (Section 3).
- The binary orientation $\sigma = \pm 1$ implies that matter and antimatter have opposite relational phase signs, leading to a repulsive gravitational force of magnitude 5/4 to 2 times Newtonian — a testable prediction (Section 4).
- σ -invariance is derived from (P1) and (P3), providing the algebraic basis for annihilation.
- The next stable level $n = 8$ ($2^8 - 1 = 255 = 3 \times 5 \times 17$) is a structurally motivated dark matter candidate. Its estimated rest-mass abundance ($\sim 0.3 M_{\text{bary}}$) falls short of observations; the dark matter problem remains open (Section 5).
- A speculative velocity synchronisation mechanism ($v \approx 0.24c$) is recorded as a structural curiosity (Section 5.4).

No experimental validation independent of the author’s prior program is claimed.

Table 1: Claim-type taxonomy.

Label	Meaning
[POSTULATE]	Foundational assumption.
[DERIVED]	Mathematical consequence of stated postulates.
[CALIBRATION]	Parameter fixed by matching one empirical value.
[HYPOTHESIS]	Testable physical conjecture.
[SPECULATIVE]	Exploratory program note; no prediction claimed.
[OPEN]	Precisely stated unresolved problem.

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1 Introduction

The relational geometry framework [1] introduces a binary orientation $\sigma = \pm 1$ in the generative algebra (Postulate 2). This orientation distinguishes left from right quaternionic action and generates the chiral decomposition $\mathfrak{so}(4) \cong \mathfrak{su}(2)_L \oplus \mathfrak{su}(2)_R$. In standard physics, matter and antimatter share the same mass, spin, and inertia but annihilate upon contact. The relational framework offers a different perspective: because σ determines the sign of the relational phase offset Φ , systems with opposite σ experience a qualitatively different gravitational interaction.

Section 2 restates the relevant postulates. Section 3 derives the propagation invariant $R_V \omega_V = c$ as a consistency condition. Section 4 derives the matter–antimatter repulsion. Section 5 analyses the $n = 8$ dark matter candidate and honestly assesses its limitations. Section 6 records a speculative bridge toward General Relativity.

2 Fundamentals of the relational framework

2.1 The generative algebra

Postulate 1 (Reference state). *Real normed vector space $\mathcal{V}$; normalised reference $\aleph$, $\|\aleph\| = 1$; differentiation marker R , $\|R\| = R \in [0, 1]$, $\langle \aleph, R \rangle = 0$. Define $\mathcal{V}_R = \{A : \langle A, R \rangle = 0\}$.*

Postulate 2 (Generative operator). *Antisymmetric bilinear $\times : \mathcal{V} \times \mathcal{V} \rightarrow \mathcal{V}$, for all $A \in \mathcal{V}_R$:*

$$\|A \times R\| = \|A\| \|R\|, \quad (\text{P1})$$

$$\langle A \times R, A \rangle = 0, \quad (\text{P2})$$

$$(A \times R) \times R = \sigma R^2 A, \quad \sigma \in \{+1, -1\}. \quad (\text{P3})$$

Postulate 3 (Time evolution). *The dynamics of a mode $A \in \mathcal{V}_R$ is governed by*

$$\frac{dA}{dt} = \omega_V A \times R,$$

where ω_V is the frequency of the V mode. Combined with (P3):

$$\frac{d^2 A}{dt^2} = \sigma R^2 \omega_V^2 A.$$

The sign of σ determines the phase orientation of the oscillation.

Postulate 4 (Orthogonal decomposition). *$U := \aleph \times R$ decomposes as $U = V + m$, $\langle m, V \rangle = 0$.*

Postulate 5 (Gravity as phase-offset reduction). *Gravity is the macroscopic tendency of systems to reduce their mutual relational phase offset $\Phi(r) = r/R_{\text{id}}$.*

Proposition 1 (Modal norms). *$\|U\| = R$, $\|V\| = R^2$, $\|m\| = R\sqrt{1 - R^2}$, $\|a\| = R^3$ where $a := V \times R$. No fifth independent mode is generated.*

Table 2: Four modes and their roles.

Mode	Norm	Role	At $n = 4$
U (arena)	R	Background, witness	ω_N
V (visible)	R^2	Relational field	$\omega_N/4$
m (confinement)	$R\sqrt{1 - R^2}$	Identity, inertia	$\sqrt{15}\omega_N/4$
a (interaction)	R^3	Adjustment	ω_N

Table 3: Closure hierarchy: arithmetic structure and physical correspondence.

n	$2^n - 1$	Factorisation	$I(n)$	Physical role
1	1	unit	1.41	Reference
2	3	prime	3.46	Electron family
3	7	prime	7.48	Neutron (unstable)
4	15	3×5	15.49	Proton (stable — ordinary matter)
5	31	prime	31.50	Supernova core (transient)
6	63	$3^2 \times 7$	63.50	Neutron star (stable collective)
7	127	Mersenne prime	127.5	Collapse threshold
8	255	$3 \times 5 \times 17$	255.5	Stable (dark matter candidate)
∞	∞	—	∞	Black hole limit

2.2 The closure hierarchy

Definition 1 (Binary closure depth and invariant). $\omega_V^2/\omega_N^2 = 2^{-n}$. $I(n) := (\omega_N/\omega_V)(\omega_m/\omega_V) = 2^{n/2}\sqrt{2^n - 1}$.

[DERIVED] $I(n)^2 = 2^n(2^n - 1) = \text{lcm}(2^n, 2^n - 1)$. For $n = 4$: $I(4)^2 = 16 \times 15 = 240$.

Structural pattern. Cases where $2^n - 1$ is prime correspond to unstable or transient states; composite cases correspond to stable configurations. This is a structural motivation, not a theorem. **[OPEN]** Derive the stability criterion from the algebraic postulates.

3 The propagation invariant as a consistency condition

[DERIVED] From the calibration $\omega_N = m_p c^2 / \hbar$, $R_V(n) = 2^{n/2} \bar{\lambda}_p$, $\omega_V(n) = \omega_N \cdot 2^{-n/2}$:

$$R_V(n) \omega_V(n) = 2^{n/2} \bar{\lambda}_p \cdot \omega_N \cdot 2^{-n/2} = \bar{\lambda}_p \omega_N = \frac{\hbar}{m_p c} \cdot \frac{m_p c^2}{\hbar} = c.$$

This holds exactly and independently of n .

Beyond the algebraic identity, there is a consistency argument:

[HYPOTHESIS] Consistency interpretation. For a closed loop in the relational network traversing coherence domains at different levels n , the accumulated phase must be well-defined. This requires a universal propagation speed c such that $R_V(n) = c \tau_V(n)$ for all n . The relation $R_V \omega_V = c$ is therefore not only a numerical consequence of the calibration but also a consistency condition for multi-scale relational closure.

Two structural consequences follow:

1. The compensation between R_V and ω_V that keeps G_{eff} nearly constant is enforced by $R_V \omega_V = c$, not accidental.
2. The relational hierarchy encodes a limiting propagation speed equal to c without assuming special relativity as a separate postulate.

4 Matter–antimatter gravitational interaction

The binary orientation $\sigma = \pm 1$ is the only discrete degree of freedom in the algebra. We identify:

- Matter: $\sigma = +1$ (left quaternionic action)
- Antimatter: $\sigma = -1$ (right quaternionic action)

Both sectors have the same modal norms and therefore the same mass, consistent with CPT symmetry.

4.1 Invariance of σ

[DERIVED] From (P1) applied twice:

$$\|(A \times R) \times R\| = \|A \times R\| \cdot \|R\| = \|A\| \cdot \|R\|^2 > 0.$$

From (P3): $\|(A \times R) \times R\| = |\sigma| R^2 \|A\|$. Therefore $|\sigma| > 0$. Since $\sigma \in \{+1, -1\}$ and cannot pass through zero, σ is a topological invariant of the domain.

[DERIVED] A relational domain cannot change σ through any continuous process allowed by the postulates. The only transformation that alters σ is the destruction of the domain. This provides the algebraic basis for matter–antimatter annihilation: when a $\sigma = +1$ and a $\sigma = -1$ domain attempt to form a joint closure, the condition $|\sigma| > 0$ for both is incompatible with a merged domain of definite σ .

4.2 Phase orientation and mutual offset

[DERIVED] From Postulate 3, the second derivative of mode A is $\sigma R^2 \omega_V^2 A$. The phase accumulated over a distance r by a domain with orientation σ is:

$$\Phi_\sigma(r) = \sigma \frac{r}{R_{\text{id}}}.$$

For two systems with opposite σ , the mutual phase offset is:

$$\Delta\Phi_{+-}(r) = \Phi_+(r) - \Phi_-(r) = \frac{r}{R_{\text{id}}} - \left(-\frac{r}{R_{\text{id}}}\right) = \frac{2r}{R_{\text{id}}}.$$

This is exactly twice the offset between two same- σ systems at the same separation.

4.3 Gravitational consequence and repulsion

[DERIVED] For two systems with the same σ , the mutual phase offset $\Delta\Phi_{++} = r/R_{\text{id}}$ is reduced by decreasing r . Postulate 5 identifies this tendency with gravitational attraction.

[HYPOTHESIS] Matter–antimatter repulsion. For two systems with opposite σ , reducing $\Delta\Phi_{+-} = 2r/R_{\text{id}}$ by decreasing r would require one domain to adopt the phase orientation of the other. But σ -invariance (derived above) forbids this without annihilation. If annihilation does not occur, the only process available is increasing r ,

which distributes the irreducible phase offset over a larger volume. This produces effective repulsion.

The step from “ σ -invariance prevents phase reduction by approach” to “the system actively increases separation” is the physical content of this hypothesis. It is consistent with the postulates but not a strict mathematical consequence of them; formalising this dynamical statement constitutes an open problem.

4.4 Magnitude of the repulsive force

The magnitude of the interaction depends on which coherence radius governs the coupling. Let R_A and R_B denote the identity radii of the two systems. The mutual phase offset gradient is:

$$\frac{d}{dr}\Delta\Phi_{+-} = \frac{1}{R_A} + \frac{1}{R_B}.$$

For same-matter reference: $\frac{d}{dr}\Delta\Phi_{++} = \frac{2}{R_A}$ (two $n = 4$ systems).

Case 1: antimatter at $n = 4$. $R_A = R_B = R_V(4)$, giving ratio = 1, force = $2GMm/r^2$.

Case 2: antimatter at $n = 8$. $R_V(8) = 4R_V(4)$, giving $\frac{1}{R_V(4)} + \frac{1}{4R_V(4)} = \frac{5}{4R_V(4)}$, force = $\frac{5}{4}GMm/r^2$.

[HYPOTHESIS] The gravitational force between ordinary matter and antimatter is repulsive with magnitude between 5/4 and 2 times Newtonian, depending on the closure level of the antimatter system. This is a falsifiable prediction.

Table 4: Antimatter gravity experiments: current status and near-future reach.

Experiment	Current result	Projected precision
ALPHA-g [2]	$g_{\text{anti}} = 0.75 \pm 0.13 g$ (downward)	$\sim 1\%$
AEgIS [3]	—	1%
GBAR [4]	—	1%

The current ALPHA-g result is consistent with downward attraction ($g_{\text{anti}} > 0$) and does not yet rule out repulsion: the prediction $g_{\text{anti}} = -1.25g$ (Case 2) lies 3.8σ from the central value, and $g_{\text{anti}} = -2g$ (Case 1) lies 10σ away, already in tension. Future measurements at 1% precision by AEgIS and GBAR will be decisive.

Falsification condition. If antimatter is confirmed to fall *downward* (attraction) at the 5σ level by any of ALPHA-g, AEgIS, or GBAR, the repulsion prediction and the σ -identification of antimatter are falsified. The framework would then require either a different identification of antimatter or a revision of Postulate 5.

5 Dark matter: $n = 8$ domains

5.1 Properties of $n = 8$ domains

[DERIVED] From the closure hierarchy:

$$\begin{aligned} R_V(8) &= 2^4 \bar{\lambda}_p = 4 R_V(4), \\ \omega_V(8) &= \omega_N/16 = \frac{1}{4} \omega_V(4), \\ \omega_m(8) &= \omega_N \cdot 2^{-4} \sqrt{255} = \frac{\sqrt{17}}{4} \omega_m(4) \approx 1.031 \omega_m(4), \\ I(8) &= 16 \sqrt{255} \approx 255.5. \end{aligned}$$

Three structural features make $n = 8$ a dark matter candidate:

1. **Same mass scale.** $\omega_m(8)/\omega_m(4) = \sqrt{17}/4 \approx 1.031$: the mass per domain is almost identical to the proton.
2. **Gravitational but optically dark.** The visible mode is suppressed by factor 4 in frequency; $n = 8$ domains do not synchronise with photons.
3. **No LCM synchronisation with ordinary matter.** The ratio $\omega_m(8)/\omega_m(4) = \sqrt{17}/4$ is irrational. The LCM period is infinite: these domains can never be at a coincidence node with $n = 4$ matter. By the WEP mechanism of the parent framework, this means they do not participate in the synchronisation that makes different materials gravitationally indistinguishable. They gravitate but do not share the WEP node with ordinary matter.

5.2 Abundance estimation

[SPECULATIVE] **Scale-linear formation hypothesis.** No derivation of $n = 8$ domain abundance exists in the framework. As a dimensional estimate only, if formation probability in the early universe is inversely proportional to coherence radius:

$$\frac{N_8}{N_4} \sim \frac{R_V(4)}{R_V(8)} = \frac{1}{4}, \quad \frac{\rho_8}{\rho_4} \sim \frac{1}{4} \cdot \frac{\sqrt{17}}{4} = \frac{\sqrt{17}}{16} \approx 0.258.$$

This is a dimensional analogy with no algebraic foundation. It is recorded here to establish an order-of-magnitude baseline, not as a prediction of the framework.

[OPEN] Derive the abundance and formation dynamics of $n = 8$ domains from the algebraic postulates or from early-universe thermodynamics.

5.3 Gravitational interaction with ordinary matter

If $n = 8$ domains have $\sigma = -1$ (opposite to ordinary matter), the mutual phase offset gradient is, from Section 4:

$$F_{48} = \frac{5}{4} \frac{GMm}{r^2}.$$

The effective mass contribution is then:

$$M_{\text{eff},8} = \frac{5}{4} M_8 \approx \frac{5}{4} \times 0.258 M_{\text{bary}} \approx 0.32 M_{\text{bary}}.$$

5.4 Velocity synchronisation as a structural curiosity

[SPECULATIVE] Velocity synchronisation note. If an $n = 8$ domain moves with velocity v such that the Lorentz factor $\gamma = 1/\sqrt{1 - v^2/c^2}$ equals the frequency ratio $\omega_m(8)/\omega_m(4) = \sqrt{17}/4 \approx 1.031$, then time dilation would exactly compensate the frequency mismatch:

$$\gamma = \frac{\omega_m(8)}{\omega_m(4)} \Rightarrow v = c\sqrt{1 - \frac{1}{\gamma^2}} = c\sqrt{1 - \frac{16}{17}} = c\sqrt{\frac{1}{17}} \approx 0.24c.$$

This observation is purely structural: the number $\sqrt{17}/4$ is small enough that its reciprocal γ yields a non-relativistic velocity ($v/c \approx 0.24$). However, the connection between the relational phase offset and the Lorentz metric has not been derived within the framework. The role of special relativity in the relational theory is not yet established. This note is recorded only as a curiosity for future work.

5.5 Assessment: why this falls short

A factor of ~ 5 additional mass is needed to explain galactic rotation curves. The framework provides at most ~ 0.32 times the baryonic mass. The shortfall is roughly one order of magnitude.

Three independent mechanisms could in principle close this gap, but none is currently derived from the framework:

- Higher-level stable domains ($n = 10, n = 12, \dots$) could contribute additional mass. Each introduces new prime factors and new irrational synchronisation ratios.
- The formation abundance could be significantly higher than the scale-linear estimate if early-universe dynamics favour larger coherence radii.
- A dimensional-transition mechanism at galactic scales could modify the effective coupling, but no such mechanism exists in the current framework.

5.6 Conclusion on dark matter

[SPECULATIVE] The relational framework structurally predicts stable $n = 8$ domains with the qualitative properties of dark matter: same mass scale as baryons, gravitationally active, optically dark, and unable to synchronise with ordinary matter via LCM. Their estimated rest-mass contribution to galactic dynamics ($\sim 0.32 M_{\text{bary}}$) falls short of observations by a factor of ~ 15 . The dark matter problem remains open within this framework.

6 A speculative bridge toward General Relativity

[SPECULATIVE] This section records a structural observation that may guide future development. It does not contain derivations.

The propagation invariant $R_V(n)\omega_V(n) = c$ implies that the coherence length R_V is the distance a relational signal travels in one oscillation period $\tau_V = 1/\omega_V$. The phase offset $\Phi(r) = r/R_{\text{id}}$ can then be rewritten as:

$$\Phi(r) = \frac{r}{c \tau_V}.$$

In a flat spacetime background, the proper time between two events separated by Δt and $\Delta \mathbf{x}$ is $\Delta\tau = \sqrt{c^2\Delta t^2 - |\Delta \mathbf{x}|^2}/c$. This suggests the identification:

$$\Phi \sim \frac{\sqrt{-g_{\mu\nu}dx^\mu dx^\nu}}{R_V},$$

where $g_{\mu\nu} = \text{diag}(-c^2, 1, 1, 1)$ is the Minkowski metric.

Caution. This identification is suggestive but circular in its current form: the Minkowski metric is being used to interpret a quantity whose geometric meaning has not yet been derived from the postulates. The relational framework derives the inverse-square law from isotropy and information conservation, but a full derivation of the spacetime metric from the postulates has not been accomplished. The covariant extension in the parent framework [1] takes the scalar field $\psi = m_p\phi_{\text{rel}}$ as a postulate, not a derivation.

The identification is recorded as a structural hint: the tendency to reduce Φ (Postulate 5) is structurally analogous to the principle of extremal proper time (geodesic motion in GR). Whether this analogy can be made precise is an open problem.

[OPEN] Derive the Minkowski interval from the relational postulates without importing the metric structure externally.

7 Testable predictions and falsification conditions

Predictions of this note:

1. **[HYPOTHESIS]** Matter–antimatter gravitational repulsion, magnitude $5/4$ to 2 times Newtonian. *Falsified if:* antimatter confirmed to fall downward at 5σ by ALPHA-g, AEGIS, or GBAR.
2. **[HYPOTHESIS]** σ -invariance implies that annihilation is the only process that can change σ . *Falsified if:* a process is found that inverts σ without domain destruction.
3. **[SPECULATIVE]** $n = 8$ domains exist and constitute a fraction ~ 0.32 of the baryonic mass contribution to gravity. *Status:* insufficient to explain rotation curves; open problem.

8 Discussion

8.1 Logical chain for antimatter repulsion

The argument for repulsion rests on four steps:

1. σ is a topological invariant of a domain **[DERIVED]** from (P1)+(P3).
2. σ determines the sign of the accumulated phase **[DERIVED]** from Postulate 3.
3. $\Delta\Phi_{+-} = 2r/R_{\text{id}}$ for opposite- σ systems **[DERIVED]** algebraically from steps 1 and 2.
4. The tendency to reduce $\Delta\Phi$ by increasing r constitutes repulsion **[HYPOTHESIS]**: the dynamical consequence of σ -invariance on Postulate 5 has not been formalised. The framework says gravity reduces $\Delta\Phi$; when this is impossible without annihilation, the repulsion is the most natural inference, but not a theorem.

[OPEN] Formalise the dynamics of Postulate 5 when $\Delta\Phi$ cannot be reduced by approach, thereby converting step 4 from **[HYPOTHESIS]** to **[DERIVED]**.

8.2 Relation to the baryon asymmetry problem

The framework begins with a reference state $\aleph$ with a fixed orientation, before the operator $\times$ is defined. This choice of $\aleph$ is not symmetric between $\sigma = +1$ and $\sigma = -1$: the reference exists prior to the distinction.

[SPECULATIVE] This asymmetry in the initial structure may provide a seed for the observed matter–antimatter asymmetry of the universe. If the reference state $\aleph$ defines a preferred orientation, $\sigma = +1$ and $\sigma = -1$ domains are not equally probable as initial conditions. This is a structural observation, not a quantitative prediction.

9 Conclusion

The relational geometry framework provides algebraic tools to address three fundamental questions:

Why does matter dominate over antimatter? The orientation σ is a topological invariant; it cannot change without annihilation. The reference state $\aleph$ has a fixed orientation, potentially providing a seed for the asymmetry. No quantitative prediction of the asymmetry ratio is made.

What is the gravitational behaviour of antimatter? The opposite sign of Φ_σ implies repulsion. The magnitude is between 5/4 and 2 times Newtonian, depending on the closure level. This is a **testable and falsifiable** prediction for ALPHA-g, AEgIS, and GBAR.

What is dark matter? The $n = 8$ level is stable, dark, massive, and unable to synchronise with ordinary matter. Its estimated contribution ($\sim 0.32 M_{\text{bary}}$) is insufficient to explain rotation curves. The dark matter problem remains open.

Summary of claim statuses:

Result	Status
σ is a topological invariant	[DERIVED]
$\Phi_\sigma = \sigma r / R_{\text{id}}$	[DERIVED]
$\Delta\Phi_{+-} = 2r / R_{\text{id}}$	[DERIVED]
Repulsion as consequence of σ -invariance	[HYPOTHESIS]
Magnitude 5/4 to 2 times Newtonian	[HYPOTHESIS]
$n = 8$ domains as dark matter candidate	[SPECULATIVE]
Abundance $\sim 0.32 M_{\text{bary}}$	[SPECULATIVE]
$\aleph$ asymmetry as seed for baryon asymmetry	[SPECULATIVE]
$R_V \omega_V = c$ as consistency condition	[HYPOTHESIS]
Bridge to GR metric	[SPECULATIVE]

A Relevant formulas from the parent framework

- $\bar{\lambda}_p = \hbar / (m_p c) \approx 2.103 \times 10^{-16} \text{ m}$
- $\omega_N = m_p c^2 / \hbar \approx 1.425 \times 10^{24} \text{ s}^{-1}$
- $R_V(n) = 2^{n/2} \bar{\lambda}_p$
- $\omega_V(n) = \omega_N \cdot 2^{-n/2}, \quad \omega_m(n) = \omega_N \cdot 2^{-n/2} \sqrt{2^n - 1}$
- $I(n) = 2^{n/2} \sqrt{2^n - 1}, \quad I(4)^2 = 240$
- $G_{\text{eff}}(n) = G \cdot \omega_m(n) / \omega_m(4), \quad \text{ceiling } \sqrt{16/15} \approx 1.033$
- $R_V(n) \omega_V(n) = c$ exactly, all n

B Open problems from this note

1. Formalise the repulsion dynamics when $\Delta\Phi$ cannot be reduced by approach.
2. Derive the stability criterion $2^n - 1$ composite $\Rightarrow$ stable from the postulates.

3. Derive the abundance and formation dynamics of $n = 8$ domains.
4. Derive the Minkowski interval from the relational postulates.
5. Determine whether the $\aleph$ asymmetry predicts a specific baryon asymmetry ratio.

LLM Disclosure Statement

Models: Anthropic Claude (primary), DeepSeek (primary theoretical stress-testing), Google Gemini (adversarial cross-validation), OpenAI ChatGPT (early versions).

Contributions: Derivation of σ -invariance from (P1)+(P3); phase-sign argument from dynamical postulate; magnitude calculation for matter–antimatter force; $n = 8$ property calculations; critical review of all claims and taxonomy labels.

Human oversight: All claims independently verified by the author. LLM suggestions that overclaimed were corrected to current labelling.

References

- [1] L. D. Mata Sánchez, *Relational Geometry and the Emergence of Gravity: From Harmonic Closure to Stellar Structure*, v13.0 (2026).
- [2] ALPHA Collaboration, *Nature* **621**, 716 (2023).
- [3] AEgIS Collaboration, arXiv:2205.02152 (2022).
- [4] GBAR Collaboration, *Hyperfine Interact.* **228**, 85 (2014).
- [5] J. H. Conway, D. A. Smith, *On Quaternions and Octonions*, A K Peters (2003).
- [6] F. Özel, P. Freire, *Annu. Rev. Astron. Astrophys.* **54**, 401 (2016).